CTFAgent: An LLM-powered Agent for CTF Challenge Solving[☆]Yuwen Zou^{a,b}, Jia Liu^{a,b}, Wenjun Fan^c ^{*}^a Xi'an Jiaotong-Liverpool University, School of Advanced Technology, 215123, China^b Department of Computer Science, University of Liverpool, L693BX, United Kingdom^c School of Engineering and Technology, University of Washington, Tacoma, 98402-3100, United States

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ABSTRACT

Capture-the-Flag (CTF) competitions play an important role in the cybersecurity landscape by simulating realistic attack and defense scenarios and offering diverse categories of challenges. This diversity demands flexible reasoning and adaptive problem-solving, which traditional automation tools struggle to provide, as they are typically designed for specific tasks. Large Language Models (LLMs) with their vast knowledge and strong reasoning capabilities, present a promising approach to overcome these limitations. In this work, we propose CTFAgent, an LLM-powered agent featuring a new plan-and-execute paradigm with a stateful task tree for long-horizon reasoning. To handle diverse challenges, CTFAgent integrates challenge-specific prompting and specialized tools for multimodal analysis and concrete operations. The agent comprises two modes: a fully automated mode and a human-in-the-loop (HITL) mode, which incorporates human operational support for tool execution beyond the automation. Evaluated on challenges from PicoCTF with GPT-4o, Gemini-2.5-Pro and DeepSeek-V3, CTFAgent outperforms 88% of human teams in its automated mode. This performance rises significantly in HITL mode, where it surpasses approximately 94% of teams. These results demonstrate that CTFAgent can effectively solve a wide range of complex tasks, highlighting the potential of LLM-powered agents to advance autonomous cybersecurity solutions.

1. Introduction

In an era of escalating digital threats, the cybersecurity landscape demands continuous innovation in both defensive and offensive capabilities. Capture-the-Flag (CTF) competitions serve as a critical benchmark for evaluating complex reasoning and multi-step problem-solving abilities essential to modern cybersecurity [1]. By simulating diverse and intricate real-world attack scenarios, these challenges require participants to integrate deep theoretical knowledge with practical hands-on skills [2]. Traditional automation tools rely on predefined rule sets that cannot adapt to novel attack patterns or unpredictable system states in dynamic environments. These tools execute tasks in isolation without considering interdependencies between attack vectors or system components. Consequently, they fail to provide the holistic planning and adaptive reasoning capabilities required for complex and multi-stage challenges.

Large Language Models (LLMs) offer a promising foundation for addressing these challenges. With extensive knowledge repositories and

emergent capabilities, LLMs can perform complex reasoning, knowledge synthesis, and domain-specific problem-solving without task-specific fine-tuning [3,4]. Recent studies demonstrate their effectiveness in cybersecurity applications, including code analysis [5], vulnerability remediation [6] and penetration testing [7,8]. Their adaptability and natural language understanding capabilities make them well suited to automate vulnerability identification, orchestrate multi-step tasks, and manage repetitive tasks, thereby enhancing both the efficiency and effectiveness of CTF challenge solving.

Preliminary studies have explored LLM-driven approaches for CTF problem-solving [9–14]. While these works demonstrate potential, they are constrained by small-scale or incomplete datasets [11], reliance on manual intervention [9,10], and limited tool integration capabilities [12]. Existing LLM agents further struggle with inherent statelessness and finite context windows when tackling complex cybersecurity tasks. These limitations underscore the need for a systematic framework integrating reasoning, planning, and tool usage, accompanied by comprehensive evaluation to assess the effectiveness of LLMs in CTF contexts.

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This paper introduces a systematic LLM-powered CTFAgent based on the plan-and-execute paradigm. Within this agent, the LLM assumes all high-level cognitive functions (including reasoning, planning, tool invocation, and evaluation), while a task tree serves as structured external memory that continuously records the agent's strategic plans and execution status of each step. This design decouples the LLM's short-term tactical reasoning from long-term strategic state management, enabling the agent to focus on concise task-relevant context at each step while maintaining continuous awareness of global objectives through the task tree. It aims to mitigate context loss and strategy drift in long-horizon tasks, ensuring behavioral coherence and effectiveness. A hierarchical prompting strategy is further employed, with high-level system prompts defining core workflows and reasoning patterns while challenge-specific prompts inject domain knowledge and problem-solving methods for different CTF categories. CTFAgent also integrates a multi-modal (e.g., source code, binaries, images and network traces) specialized tool suite, enabling interaction with real environments through concrete operations such as file analysis, network probing and code debugging, thereby breaking the limitations of pure text-based reasoning.

CTFAgent supports two operation modes: a fully automated mode for evaluating autonomous problem-solving capabilities and a human-in-the-loop (HITL) mode that provides operational support only when automated tool execution fails. This mode decouples the LLM's core reasoning capabilities from tool execution errors, enabling fair assessment of the model's intrinsic abilities. The agent implementation has been made publicly available on GitHub.¹

The contributions of this paper are summarized as follows:

- An LLM-powered CTFAgent has been proposed with two modes: automated and HITL. In the HITL mode, human operators only provide operational support without directing LLM's strategy.
- CTFAgent creates and applies a task tree for long-horizon reasoning, dynamically generating tailored prompts and invoking multi-modal specialized tools to solve diverse challenges.
- A comprehensive evaluation of the CTFAgent has been conducted, which leverages three LLMs (GPT-4o, Gemini-2.5-Pro, and DeepSeek-V3), demonstrating the effectiveness and generalizability of the agent.
- The HITL mode in CTFAgent is used to address the limitation of the multi-modal specialized tools invocation, and so stress on evaluating the core reasoning of LLMs, revealing their performance boundaries and intrinsic limitations on CTF challenges.

The remainder of this paper is organized as follows: Section 2 presents a comprehensive survey of LLM agents and LLM-guided CTF research. Section 3 elaborates on the design of the LLM-powered CTFAgent. Section 4 details the prototype implementation. Section 5 presents experimental results. Finally, Section 6 summarizes key findings and outlines future research directions.

2. Related work

This section reviews recent literature on the applications of LLMs in cybersecurity, with particular attention to their emerging role in CTF challenges.

2.1. LLM applications in cybersecurity

The integration of LLMs into cybersecurity has enabled notable advancements in defensive and offensive capabilities. Research primarily addresses three domains: threat detection, automated security testing, and collaborative agent systems.

In threat detection and deception, LLMs demonstrate enhanced capability to analyze complex security threats. Alzaabi et al. [15] emphasized the challenges and intricacies of insider threat detection, noting that while classical machine learning methods are widely adopted, they often fall short in capturing the nuanced behaviors of insiders. This has led researchers to explore LLMs as alternatives capable of handling these complexities, offering promising new ways to detect and analyze insider threat activities. Hossain [16] proposed a novel approach using LLMs to enhance malicious code detection in Java source code via the Mixtral architecture, which outperformed existing static analysis tools in identifying malicious code. Guo et al. [17] examined the effectiveness of LLMs in detecting software vulnerabilities, highlighting the impact of fine-tuning on classification accuracy and raising concerns about mislabeling in benchmark datasets. Additionally, LLMs have demonstrated potential in creating realistic honeypots that can deceive and engage attackers. For example, Sladic et al. [18] introduced shellLM, a honeypot leveraging LLMs to produce realistic Linux-like shell interactions, achieving a 90% success rate in deceiving cybersecurity experts through advanced prompt engineering. Similarly, Fan et al. [19] proposed HoneyLLM, a medium-interaction honeypot that uses LLMs to generate realistic shell responses, capturing detailed attacker behaviors while minimizing the risks associated with high-interaction systems. These studies collectively suggest that LLMs can significantly enhance detection and deception capabilities in cybersecurity.

For automated vulnerability assessment and penetration testing, LLMs serve as tools for identifying and exploiting system weaknesses. Bianou et al. [20] presented the PENTEST-AI framework, which automates penetration testing through LLMs, with specialized agents responsible for scanning, vulnerability validation and reporting. In addition, Fang et al. [21] have shown that LLMs, particularly GPT-4, can autonomously exploit one-day vulnerabilities, achieving up to an 87% success rate for critical vulnerabilities when provided with CVE descriptions. Huang and Zhu [22] introduced PenHeal, a two-stage LLM-based framework that autonomously identified and mitigated vulnerabilities, achieving better coverage and remediation compared to baseline models.

Multi-agent collaboration and human-agent interaction further enhance the effectiveness of LLM applications. Qian et al. [23] demonstrated the effectiveness of scaling multi-agent networks in LLM applications, showing how collaborative intelligence among agents can surpass individual capabilities, thus streamlining interactive reasoning processes. Feng et al. [24] delved into the realm of LLM-based human-agent collaboration for complex task-solving, showcasing the synergistic potential of such collaborations in offline reinforcement learning environments. Strategic prompt engineering has also been identified as critical for optimizing LLM-based defense mechanisms [25].

Overall, these works establish LLMs as versatile tools for diverse cybersecurity challenges. Their capabilities in threat detection, automated exploitation and collaborative problem-solving provide a foundation for applications in complex environments like CTF competitions.

2.2. LLM applications in CTF

CTF competitions serve as standardized environments for evaluating and honing cybersecurity skills, making them ideal benchmarks for assessing the capabilities of AI agents [1,26]. Consequently, research has increasingly focused on applying LLMs to solve CTF challenges.

Initial explorations involved manual evaluations where models like ChatGPT, Google Bard and Microsoft Bing were tested against various CTF problems with human assistance [9,10]. While these studies provided preliminary insights, their scope was limited. Subsequent research has shifted toward fully automated agents. AutoAttacker [27] introduced a benchmark to evaluate LLM-based attack automation across various difficulty levels. Abramovich et al. [28] presented EnIGMA,

¹ <https://github.com/Zoey-oon/CTFLLM>.

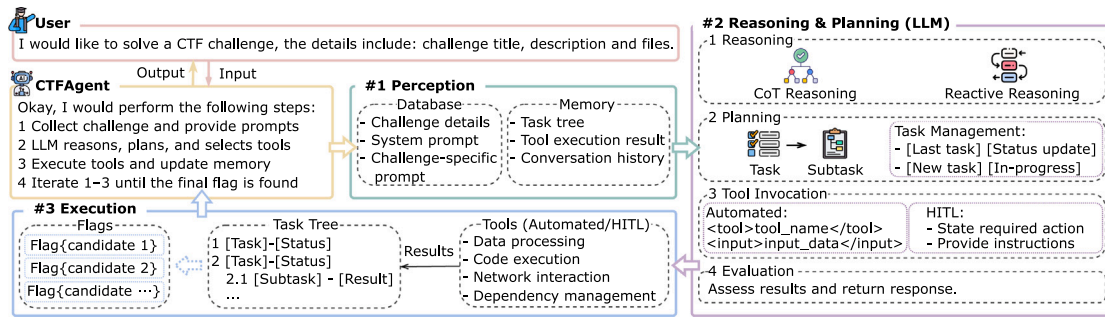


Fig. 1. The LLM-based CTFAgent, comprising two modes: automated and HITL (the latter with human operational support for tool execution beyond automation).

an agent that autonomously tackles CTF tasks using innovative Agent-Computer Interfaces (ACIs) to improve success rates, achieving top scores on benchmarks like NYU CTF and Intercode-CTF. Laney [29] further explored LLMs' automation abilities, achieving a notable improvement in PicoCTF task completion and proposing a hierarchical framework for structured task handling in complex challenges.

Researchers have also investigated optimal workflows for LLM-based CTF solving. Shao et al. [11] demonstrated that LLMs could perform similarly to humans on CTF tasks when provided with relevant information, but noted limitations in current models, such as tool selection and task-specific functionalities. An extension of this work [12] introduced a public dataset and an automated solving framework but demonstrated deficiencies in tool integration and process balance, suggesting that LLM agents might not yet fully match human adaptability in all scenarios.

Despite significant progress in automating CTF solutions, key challenges remain in tool selection, process accuracy and the development of balanced evaluation metrics. This work aims to address these limitations by developing a more robust framework to optimize LLM performance in complex real-world cybersecurity tasks.

3. Design of CTFAgent

This section outlines the design of LLM-based CTFAgent with key components.

3.1. Agent overview

The LLM-based CTFAgent is engineered to address CTF challenges and supports two operational modes: a fully automated mode and a human-in-the-loop (HITL) mode, as illustrated in Fig. 1. Across both operational modes, the agent adheres to a core iterative workflow. This cycle commences with perception, wherein the agent gathers all available challenge information (e.g., descriptions, files, URLs) and structures it into a formatted prompt. Subsequently, during reasoning and planning, the LLM processes this prompt to analyze the problem, formulate a strategy and generate tool commands. In this stage, the LLM performs four steps: reasoning, planning, tool invocation and evaluation. During the reasoning step, the LLM analyzes the challenge context and task progress to clarify the requirements, applying both Chain-of-Thought (CoT) reasoning for systematic analysis and reactive reasoning for dynamic adaptation. In the planning step, the LLM updates the task tree to reflect the outcome of the previous step and formulates the next strategy. In the tool invocation step, the LLM selects appropriate tools and provides execution parameters. In the evaluation step, which concludes the reasoning and planning stage, the LLM applies pattern matching to detect potential flags and determines whether further iterations are required. In some cases, evaluation may only require the LLM to produce the final flag in the correct format without invoking any tools, particularly when sufficient information is already available. The execution stage then runs these commands

against the challenge environment, generating results that are stored in memory and subsequently fed back into the workflow.

This process follows a CoT reasoning approach in which each step builds on the previous one to maintain logical consistency and enable deeper analysis. The agent executes only one step per response, with each iteration reflecting a single reasoning step, followed by tool execution and task tree updates. If no flag is detected, the process continues with updated task information. A maximum iteration limit is enforced to prevent infinite loops and ensure efficiency. The agent also reflects the principles of Reasoning and Acting (ReAct), where reasoning and tool execution alternate throughout the workflow. This iterative approach allows the agent to adapt to changes in the task tree and ensures efficient solutions to CTF challenges.

3.1.1. Automated mode

In the fully automated mode, the agent executes the core workflow autonomously. The system directly invokes the tools selected by the LLM and feeds the output back into the loop without any human intervention. This mode is suitable for challenges where the environment and tools are fully compatible with automated execution.

3.1.2. HITL mode

The HITL mode maintains the fundamental architecture of the automated mode, differing solely in the potential for manual tool execution when automation proves infeasible. Human intervention in this setting is strictly limited to operational support, defined as the manual execution of LLM-generated tool commands exactly as instructed. The operator may re-run failed commands or handle basic input/output operations; however, they are explicitly prohibited from modifying, debugging, or reinterpreting the commands. This restriction ensures reproducibility by excluding any intellectual contribution beyond the LLM's original outputs. In this mode, the LLM retains full responsibility for reasoning, planning, and generating commands, while the human operator intervenes only to perform actions that cannot be automated due to security restrictions, reliance on external resources, or the need for manual interaction. This division of labor allows the agent to circumvent the limitations of full automation, thereby enhancing its operational reliability and success rate in complex environments where automated tool use is constrained.

3.2. Prompt engineering

Prompt engineering provides the control framework that constrains the behavior of the CTFAgent in CTF solving. It enforces structured workflows for reasoning, action and task management, ensuring systematic problem decomposition and execution under the principles of CoT reasoning and the ReAct paradigm. The framework is composed of a system prompt that defines global behavioral rules and challenge-specific prompts that adapt the workflow to individual tasks.

Table 1
Classification of CTF challenges.

Category	Abbrev.	Description
Web Exploitation	Web	Exploit web vulnerabilities such as XSS, CSRF, injection, misconfigurations and logic flaws
Cryptography	Crypt	Break or analyze cryptographic schemes through design flaws or weak implementations
Reverse Engineering	Rev	Analyze binaries using disassembly or debugging to recover program logic, algorithms or hidden data
Forensics	Foren	Examine memory, network, or file system artifacts to extract digital evidence or hidden information
General Skills	GS	Apply foundational skills such as scripting, file manipulation, and command-line utilities
Binary Exploitation	Pwn	Exploit binary vulnerabilities to gain control, bypass protections or escalate privileges

Table 2
Tool integration design for the CTFAgent.

Category	Tool	Description
Data and file interaction	Zip Handler	List and extract contents from compressed ZIP archives
	File Reader	Read and inspect the contents of arbitrary files
	QR Reader	Decode and extract embedded data from QR code images
Code and command execution	Package Installer	Dynamically install required Python packages to adapt to unforeseen requirements
	System Command	Execute essential shell commands from a curated security whitelist
	Code Executor	Execute Python scripts for custom logic
	JS Executor	Execute JavaScript code essential for web-based challenges
Network interaction	Network Connector	Establish and interacts with remote services via TCP sockets
Agent control	Task Manager	Provide an interface for the LLM to programmatically update the task tree
	Flag Validator	Submit a potential flag for validation through the designated service

Role: You are an expert CTF challenge solver with strong analytical and problem-solving skills.	
Global Rules: - Goal: Find the flag → final format xxx{...}. - Loop: One step per response (Reason → Act). - Reasoning: Always precede tool usage. - Tool usage: Only for new info/actions. - Tool Limit: Max 3 tools per response. - Response structure: Reasoning→Planning→Tool Invocation→Evaluation	Task Management: - Maintain a task tree with updates. - Format: Task [number]. [description] - [status] - One update per step. - Status: pending/in-progress/completed/failed.
Available Tools: - file_reader: Read files from challenge directory - zip_handler: Handle ZIP file operations - code_executor: Execute Python code (include SSH) - system_command: Execute system commands - network_connector: Connect to remote services - js_executor: Execute JavaScript code - qr_code_reader: Read QR codes from images - package_installer: Install Python packages - task_manager: Manage task tree - flag_validator: Extract and validate flags	Tool Reliability: - If a tool fails (command not found, timeout, etc.), adapt your approach. - Use alternative tools or methods when primary tools are unavailable. - Don't let tool failures block your exploitation strategy. - Consider manual execution when tools fail.
Strategic Workflow (Adaptive): - Reason about challenge and available information. - Update task status in the required format. - Execute tools only when necessary. - If code provided, analyze directly. - If tools unavailable, use alternative methods. - Avoid redundant information gathering. - Assess results and determine the next step.	Mode Guidelines: - Automated mode: Agent executes tools automatically. - HITL mode: If tool not executable, state exact manual command for human.

Fig. 2. System prompt template.

3.2.1. System prompt

The system prompt imposes a deterministic and verifiable control framework on the LLM's inherently non-deterministic behavior. As detailed in Fig. 2, the prompt transforms the general-purpose model into a specialized agent by enforcing a structured problem-solving loop. This loop requires that each action be preceded by explicit analysis, organized into a reasoning, planning, tool invocation and evaluation sequence, thereby instantiating the ReAct paradigm. To maintain coherence across multiple steps, the prompt establishes task management rules that oblige the agent to preserve a structured task tree, serving as its dynamic state representation and procedural plan.

In addition to this analytical framework, the prompt defines the agent's operational capabilities and its rules for adaptive behavior. It specifies a bounded set of tools and provides explicit procedures for handling tool failures, ensuring operational reliability. The workflow guidelines further instruct the agent to act with deliberation, avoiding redundant actions and executing tools only when necessary. Finally, the mode guidelines set out a protocol for human collaboration in HITL mode. Overall, these instructions form a comprehensive operational framework that governs all aspects of the agent's behavior.

3.2.2. Challenge-specific prompt

To optimize agent performance, a tailored prompt is created for each CTF category because these challenges require unique domain knowledge. As detailed in Table 1, the domains are highly specialized. Web exploitation addresses application-layer vulnerabilities and cryptography focuses on analyzing flawed cryptographic schemes. Reverse engineering emphasizes deconstructing program logic, forensics involves examining digital artifacts, and binary exploitation demands manipulating low-level memory structures. By providing targeted contextual guidance, the framework enhances the agent's adaptability and enables it to execute specialized strategies appropriate for each category.

This guidance is dynamically generated for each challenge by selecting a predefined template based on its category metadata. As illustrated in Fig. 3, each template is a structured guide that provides the LLM with illustrative domain-specific information. The relevant knowledge section serves as a reference for key concepts while the typical approaches section suggests common analytical workflows. Further sections enumerate common tools as examples and highlight common exploitation patterns to provide strategic heuristics. This information is appended after the specific challenge description to form a complete context. This method primes the LLM with expert-level context, channeling its reasoning toward effective strategies without strictly constraining its problem-solving autonomy.

3.3. Task management

The CTFAgent uses a task management mechanism to structure its stateful multi-step problem-solving process. This mechanism is centered on a task tree, a hierarchical data structure that maintains a dynamic representation of the agent's evolving strategic plan and its execution status. The task tree externalizes the agent's reasoning process, allowing for the decomposition of a complex challenge into a structured hierarchy of sub-tasks. This approach ensures long-term strategic coherence and addresses the inherent limitation of LLMs to focus only on the most recent interaction.

The operational cycle unfolds over a series of structured updates to the task tree with a clear division of labor. In each cycle, the LLM acts as the strategic planner, analyzing the entire tree to generate natural language directives for its modification. These directives specify actions such as modifying task statuses, creating new tasks, or decomposing existing tasks into new sub-tasks. The agent's framework then parses

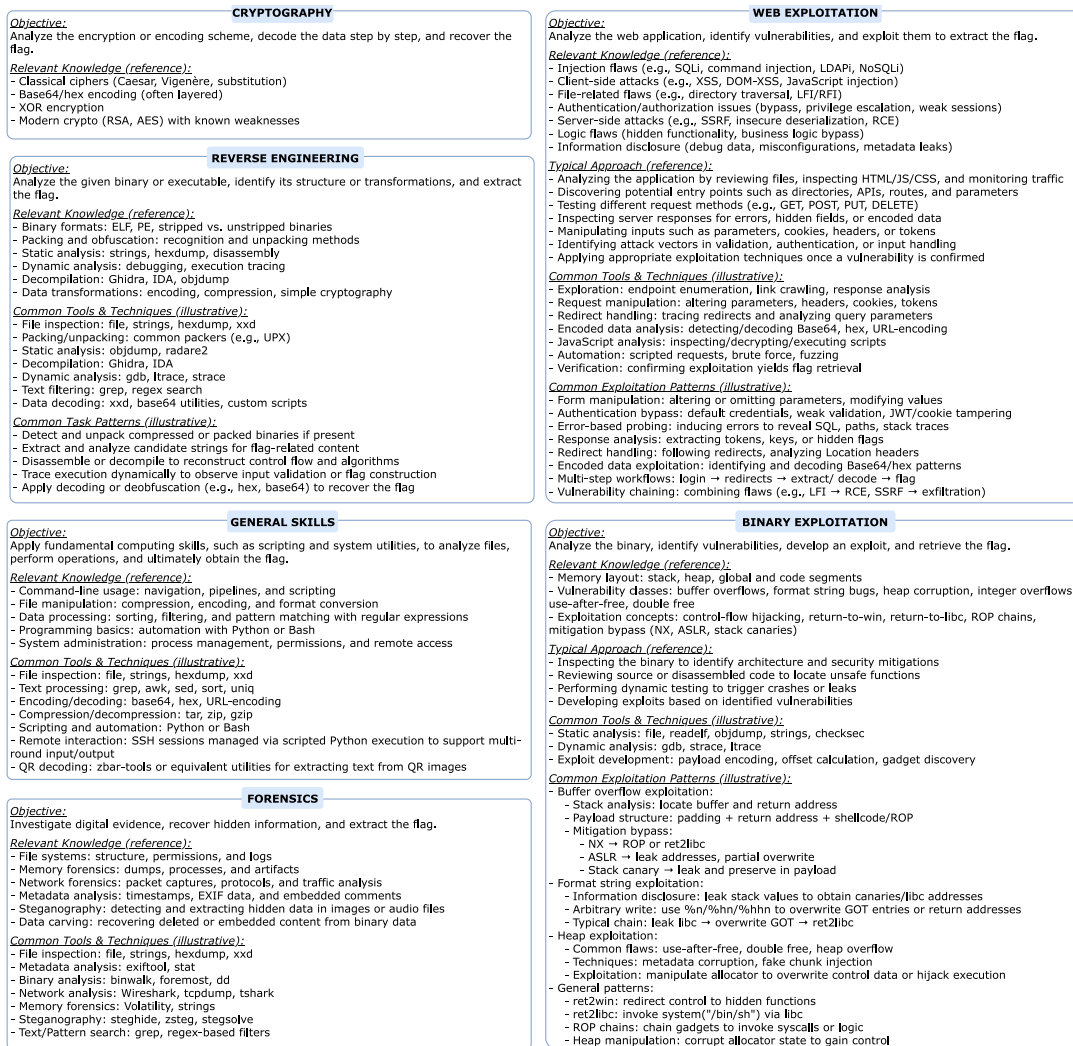


Fig. 3. Challenge-specific prompt templates.

these directives. This parsing step is critical as it validates the proposed modifications and safeguards the structural integrity of the plan against potential LLM hallucinations. Once validated, the agent updates the tree, executes the corresponding tool, and programmatically associates the outcome with the relevant task node. The fully updated tree is then returned to the LLM as the context for the subsequent reasoning cycle. For operational resilience, this entire state is serialized to persistent storage, ensuring robust and restorable long-term planning capabilities.

3.4. Tool integration

The tool integration layer serves as the critical interface between the LLM's abstract reasoning and the concrete execution environment. It translates the planner's high-level directives into deterministic and verifiable actions, enabling the agent to interact with the challenge environment. The toolset is designed to be comprehensive, providing a sufficient range of capabilities while mitigating the risks of open-ended code execution. Each tool is implemented as a discrete sandboxed function with a strictly defined interface to ensure that the LLM's invocations are both safe and reliable.

The available tools are organized into four functional categories to provide a clear operational structure, with specific tools and their descriptions detailed in Table 2. Data and File Interaction tools are fundamental for perception, providing utilities to inspect files, handle archives and decode specialized data formats like QR codes. Code and

Command Execution is facilitated through controlled sandboxed environments, including access to a secure whitelist of shell commands and interpreters for Python and JavaScript. Network Interaction is handled by a dedicated tool for communicating with remote services, which is indispensable for many challenges. Finally, Agent Control is managed by tools that operate on the agent's meta-level, allowing it to manage its task plan and validate the final flag.

4. Implementation

This section details the implementation of the CTFagent.

4.1. System environment

The CTFagent is designed to operate within a Kali Linux environment, it is the industry-standard distribution for penetration testing and provides a comprehensive, pre-configured suite of the specialized security tools and libraries that the agent's toolset requires. To ensure this specialized environment is portable across diverse host systems and to guarantee a standardized and reproducible setup, the entire agent and its required Kali Linux environment are encapsulated within a Docker container. This containerized approach provides a consistent and isolated execution environment fundamental for experimental validity.

Table 3
Implementation of the CTFAgent's toolset.

Tool	Implementation (Commands & Python libraries)
Zip Handler	Python libraries: zipfile, tarfile, gzip System commands: unzip, tar, gzip, bzip2
File Reader QR Reader	System commands: cat, head, tail, less, file, stat, pwd, find, grep, strings, wc, sort, uniq, cut, awk, sed Python libraries: pyzbar, opencv-python, Pillow
Code Executor	A Python REPL session managed by the pexpect library. Enables runtime use of Python libraries: pwntools, requests, socket, paramiko, BeautifulSoup4, urllib, json, re, etc
JS Executor Package Installer	A nodejs subprocess for executing JavaScript snippets subprocess.run calls to package managers: pip install, apt install, apt-get install
System Command	A secure subprocess.run wrapper for commands including: ls, cd, xxd, hexdump, checksec, foremost, binwalk, steghide, zsteg, exiftool, git, gdb, nmap, netcat, curl, gobuster, dirb, sqlmap, tshark, openssl, hashcat, john, wget, ssh, ping
Network Connector	Primarily low-level TCP/IP interaction via Python's socket library and the netcat command. High-level requests are typically delegated to Python libraries like requests within the Code Executor
Task Manager	In-memory Python class (TaskTree) using list/dict structures, with methods for state management, LLM response parsing (re module), and JSON serialization for persistence
Flag Validator	Python's re module for local pattern matching against the standard format

Table 4
Key operational parameters.

Category	Parameter	Value
LLM inference	LLM temperature	0.7
Execution constraint	Maximum interaction rounds	30
Context management	Context token limit	120,000 tokens
	Recent task context size	3 tasks
	Full context interval	5 rounds

4.2. Agent orchestration

The agent's operation is orchestrated by a custom engine that directs the entire problem-solving lifecycle, leveraging the LangChain framework for its core reasoning capabilities. Each operational cycle begins with our engine performing context management to select and prepare the relevant history for the LLM. It then delegates a single-step reasoning task to LangChain's agent executor, which implements the core ReAct logic: the LLM generates a thought and an action, which is a tool invocation. Our engine then resumes control, intercepting this action. It parses the LLM's response, records the tool's outcome and updates the persistent task tree, thereby completing the cycle. This hybrid architecture places the strategic and stateful control logic within our custom implementation while using LangChain as a powerful but contained engine for individual reasoning steps.

4.3. Tool implementation

As summarized in Table 3, the agent's tools are realized through a combination of direct command-line wrappers and custom Python functions. The agent leverages the rich ecosystem of the Kali Linux environment by invoking standard uses (e.g., nmap, sqlmap) through the system command interface. For tasks requiring more complex logic, tools are implemented as dedicated Python functions that utilize specialized libraries (e.g., pwntools, requests) and are typically accessed via the code executor interface. The following table provides a definitive mapping of the agent's tools to their underlying implementation details.

5. Experiments

This section evaluates the performance of CTFAgent in automated and HITL modes, comparing their effectiveness in solving CTF challenges.

5.1. Experimental setup

The performance of the CTFAgent is evaluated using three LLMs, selected for a comparative analysis across key variables. OpenAI's GPT-4o serves as the general-purpose model, Google's Gemini-2.5-Pro provides a large context window, and the open-source DeepSeek-V3 represents models specifically optimized for code and technical reasoning. To ensure a fair and consistent comparison of the models' core reasoning abilities, a fixed set of operational parameters was used across all experiments. These parameters, which govern the agent's execution constraints and context management, are detailed in Table 4.

During experiments, LLMs are not allowed to search the internet for solutions or external references. Solutions are generated solely based on the models' internal knowledge base, ensuring that results reflect the models' inherent problem-solving abilities without external aid.

5.2. Benchmark dataset

The benchmark dataset for this study is composed of challenges from the PicoCTF platform,² a widely recognized cybersecurity competition suitable for systematic analysis. The selected tasks are drawn from the 2023 and 2024 competitions, with a comprehensive list provided in Table 5. For the purpose of our analysis, these challenges are categorized by their official domain (i.e., web exploitation, cryptography, reverse engineering, forensics, general skills, and binary exploitation) and difficulty level (i.e., easy, medium, hard).

Three challenges (i.e., *who is it*, *money-ware* and *permissions*) from the PicoCTF 2023 competition were unavailable on the platform at the time of our experiments. Consequently, these three challenges are excluded from our success rate statistics (i.e., they are not included in the denominator N_{total}). However, for the competitive ranking analysis against other teams in that year, CTFAgent is considered to have scored zero points on these tasks.

The distribution of these challenges is visualized in Figs. 4 and 5. In each figure, challenges are arranged by category and their easy, medium, or hard difficulty labels on the x -axis. The height of each bar represents the number of public solves (collected from the PicoCTF website on October 29, 2024), while the color indicates the assigned point value. These figures highlight a crucial distinction between two types of difficulty in the dataset. The first is the pre-defined difficulty, represented by the point values assigned by challenge creators. The second is the empirical difficulty, which is represented by the easy, medium and hard labels provided by the PicoCTF platform

² <https://picoctf.org>.

Table 5
Challenges in PicoCTF 2023 and 2024 (Ordered by category and solve count).

PicoCTF 2023											
ID	Challenge	ID	Challenge	ID	Challenge	ID	Challenge	ID	Challenge	ID	Challenge
C1	findme	C8	rotation	C15	Reverse	C22	No way out	C29	UnforgottenBits	C36	two-sum
C2	MatchTheRegex	C9	ReadMyCert	C16	Safe Opener 2	C23	Virtual Machine 1	C30	repetitions	C37	babygame01
C3	SOAP	C10	HideToSee	C17	timer	C24	PcapPoisoning	C31	chrono	C38	VNE
C4	More SQLi	C11	SRA	C18	Ready Gladiator 0	C25	hideme	C32	Permissions	C39	hijacking
C5	Java Code Analysis!?!	C12	PowerAnalysis: Warmup	C19	Ready Gladiator 1	C26	FindAndOpen	C33	useless	C40	tic-tac
C6	msfroggenerator2	C13	PowerAnalysis: Part 2	C20	Virtual Machine 0	C27	MSB	C34	Special	C41	babygame02
C7	cancri-sp	C14	PowerAnalysis: Part 1	C21	Ready Gladiator 2	C28	Invisible WORDs	C35	Specialer	C42	Horsetrack
PicoCTF 2024											
ID	Challenge	ID	Challenge	ID	Challenge	ID	Challenge	ID	Challenge	ID	Challenge
C43	WebDecode	C51	Custom encryption	C59	weirdSnake	C67	endianness-v2	C75	Collaborative Development	C83	format string 1
C44	Unminify	C52	C3	C60	WinAntiDbg0x200	C68	Blast from the past	C76	binhexa	C84	heap 2
C45	Bookmarklet	C53	rsa_oracle	C61	WinAntiDbg0x300	C69	Dear Diary	C77	endianness	C85	heap 3
C46	IntroToBurp	C54	flag_printer	C62	Scan Surprise	C70	Super SSH	C78	dont-you-love-banners	C86	format string 2
C47	Trickster	C55	packer	C63	Verify	C71	Binary Search	C79	SansAlpha	C87	format string 3
C48	No Sql Injection	C56	FactCheck	C64	Secret of the Polyglot	C72	Time Machine	C80	heap 0	C88	babygame03
C49	elements	C57	WinAntiDbg0x100	C65	CanYouSee	C73	Commitment Issues	C81	format string 0	C89	high frequency troubles
C50	interencdec	C58	Classic Crackme 0x100	C66	Mob psycho	C74	Blame Game	C82	heap 1		

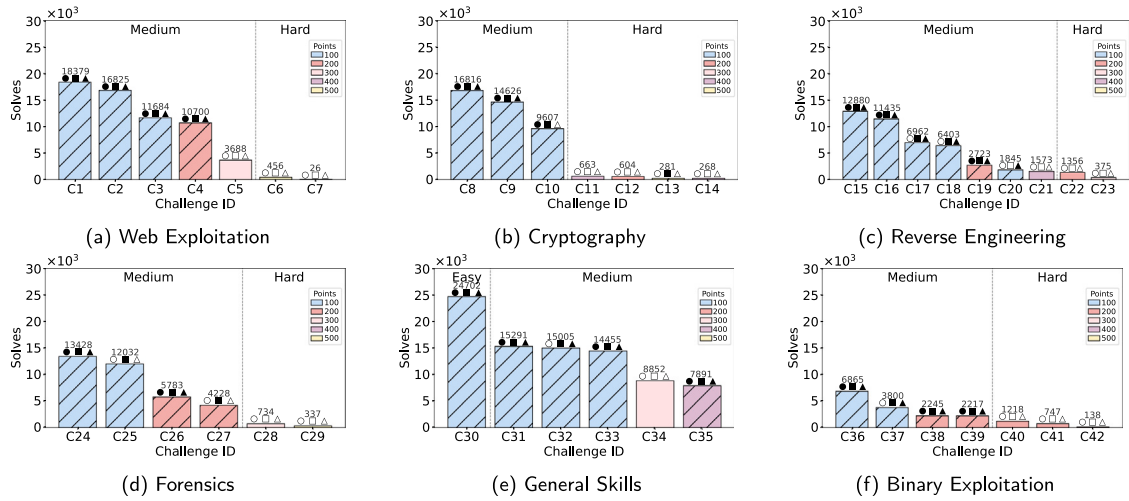


Fig. 4. Performance of CTFAgent with different LLM backbones on PicoCTF 2023 challenges (Agent solve status is marked by model: GPT-4o (●/○), Gemini-2.5-Pro (■/□), and DeepSeek-V3 (▲/△), where solid and hollow indicates solved and unsolved respectively. Hatched bars denote challenges solved by exactly one model). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

and is based on public solve counts. A key characteristic revealed in the figures is that these two measures are not always aligned, a challenge's point value often does not directly correlate with its public solve count. This discrepancy underscores the difference between the intended complexity and the actual difficulty perceived by the community.

5.3. Overall performance evaluation

To systematically evaluate the problem-solving capabilities of the LLM agents, we adopt the success rate as our primary metric. The success rate is defined as the ratio of challenges successfully solved to the total number of challenges within a given subset:

$$\text{Success Rate} = \frac{N_{\text{solved}}}{N_{\text{total}}} \times 100\% \quad (1)$$

This metric provides a standardized measure for comparing performance across different LLMs, operational modes and challenge granularities. To ensure a fair and unbiased evaluation, we verified that the LLM backbones had no prior exposure to the specific challenges used in benchmark dataset.

Figs. 4 and 5 present a detailed per-challenge breakdown of agent performance. The agent's solve status under different LLM configurations is denoted by symbols, with a solid symbol for a solved challenge and a hollow one for an unsolved one. Hatched bars indicate challenges solved by exactly one agent. An initial observation confirms the expected trend: successful solves are predominantly concentrated in challenges designated as easy or medium. However, a deeper analysis reveals a more noteworthy finding regarding the perception of difficulty. While a discrepancy often exists between a challenge's pre-defined difficulty and its empirical difficulty for humans, we found that the LLM agent's performance aligns remarkably well with these difficulty. Specifically, there is a strong inverse correlation where the agent's success rate predictably decreases as challenge points increase. We hypothesize this is because the LLM excels at problems with structured and scalable complexity that the point system accurately reflects. This contrasts sharply with human performance, which is often influenced by unpredictable factors like unique insights or specialized knowledge, leading to results that can deviate from the expected difficulty curve.

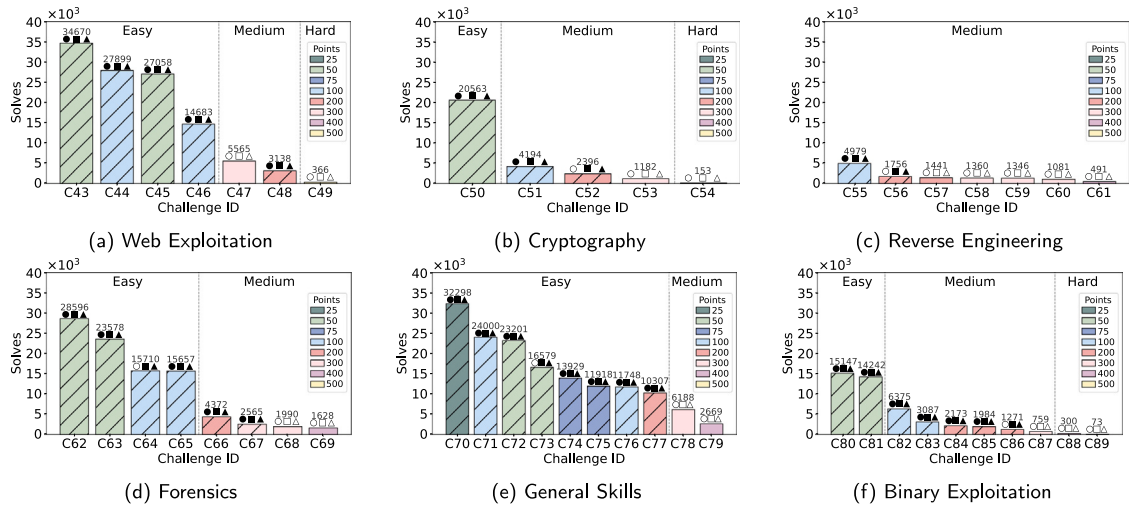


Fig. 5. Performance of CTFAgent with different LLM backbones on PicoCTF 2024 challenges (Agent solve status is marked by model: GPT-4o (●/○), Gemini-2.5-Pro (■/□), and DeepSeek-V3 (▲/△), where solid and hollow icons indicate solved and unsolved challenges, respectively. Hatched bars denote challenges solved by exactly one model). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

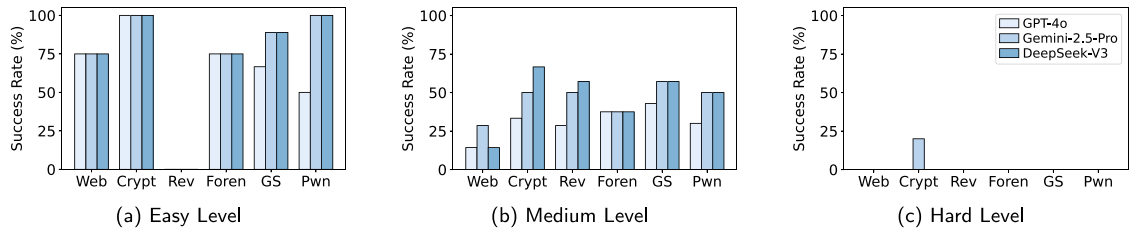


Fig. 6. Comparison of different LLM success rates in CTFAgent's automated mode on PicoCTF 2023–2024 challenges.

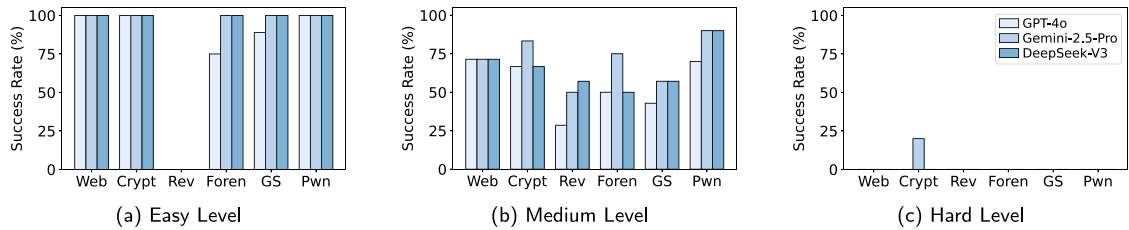


Fig. 7. Comparison of different LLM success rates in CTFAgent's HITL mode on PicoCTF 2023–2024 challenges.

Finding 1: LLM-based agents exhibit a more predictable relationship between performance and pre-defined difficulty than human solvers, whose success is often influenced by non-linear strategies and insights.

To quantify these trends, Figs. 6 and 7 aggregate these results into success rates for the automated and HITL modes, respectively. In the automated mode, performance on easy challenges is generally high. With the exception of the reverse engineering category, which contains no easy challenges in our dataset, all models achieve excellent results on the remaining easy tasks. However, this success declines sharply for medium challenges. For instance, in the medium cryptography category, DeepSeek-V3 achieved a 66.7% success rate, followed by Gemini-2.5-Pro at 50%, while GPT-4o managed only 33.3%. Hard challenges prove almost entirely intractable in this mode, with Gemini-2.5-Pro being the only agent to solve a single hard challenge in cryptography. Agent performance is also highly domain-specific, with the highest success rates in cryptography, and the lowest in reverse engineering, which require a deeper understanding of low-level program states.

Fig. 7 presents the success rates in the HITL mode. This mode is designed to provide a more accurate evaluation of an agent's core

reasoning capabilities by using a human operator to bypass environmental factors and potential execution-level failures. This allows us to measure the model's performance when the reliability of command execution is guaranteed. A substantial performance improvement is observed under these controlled conditions, particularly on medium challenges. For instance, in the binary exploitation category, Gemini-2.5-Pro's success rate jumps from 50% in the automated mode to 90% in the HITL mode. GPT-4o's performance is also dramatically elevated. Its success rate in the medium web category surged from a mere 28.6% to 71.4%. This wide gap suggests that the results from the automated mode may underestimate the models' true problem-solving potential. The underlying reasoning and generated plans are often sound, but their success can be masked by the inherent brittleness of a fully automated execution environment. Despite these significant gains on medium tasks, hard challenges remain almost entirely unsolvable. The lack of improvement in HITL mode indicates that for these problems, the primary bottleneck is indeed the LLM's core reasoning and planning capabilities, not the reliability of execution.

Finding 2: Agents show potential in solving simple CTF challenges, but they struggle with more difficult problems.

Table 6

Performance comparison of CTFAgent and human teams in PicoCTF 2023 and 2024.

Year	Model	Mode	LLM score	Total score	Ranking (%)
2023	GPT-4o	Automated	1800	9100	18.27
		HITL	2700		11.17
	Gemini-2.5-Pro	Automated	2600		<u>11.71</u>
		HITL	3900		<u>6.11</u>
	DeepSeek-V3	Automated	2200		14.59
		HITL	3100		9.17
2024	GPT-4o	Automated	1325	9225	21.32
		HITL	2775		8.47
	Gemini-2.5-Pro	Automated	2525		<u>9.63</u>
		HITL	3525		<u>6.05</u>
	DeepSeek-V3	Automated	2525		<u>9.63</u>
		HITL	3525		<u>6.05</u>

Table 7

Success rates per difficulty levels in Single-Step (S) vs. Multi-Step (M) tasks.

Difficulty Step	Easy		Medium		Hard	
	S	M	S	M	S	M
Web	100.00%	/	100.00%	33.33%	/	0.00%
Crypt	100.00%	/	100.00%	66.67%	/	20.00%
Rev	/	/	100.00%	45.45%	/	0.00%
Foren	100.00%	100.00%	/	75.00%	/	0.00%
GS	100.00%	100.00%	/	57.14%	/	/
Pwn	/	100.00%	66.67%	85.71%	/	0.00%

Table 6 evaluates the competitive performance of the CTFAgent against a field of over 10,000 teams in the PicoCTF 2023 and 2024 competitions. The agent, powered by three different LLMs in both automated and HITL modes, is measured by its final rank as a percentage of all participants, where a lower value indicates a superior result. The results demonstrate the agent’s high competitiveness in both operational modes. Even when operating fully autonomously, the agent performed impressively, with Gemini-2.5-Pro ranking in the top 11.71% in 2023, and both Gemini-2.5-Pro and DeepSeek-V3 tying for a rank in the top 9.63% in 2024. Performance was significantly elevated in the HITL mode, yielding the best overall results: Gemini-2.5-Pro achieved a top 6.11% placement in 2023, and tied with DeepSeek-V3 for the top 6.05% rank in 2024. This firmly establishes the CTFAgent as a competitor with the leading HITL configuration outperforming approximately 94% of all teams.

5.4. Task horizon analysis

To understand how the number of required steps impacts performance, we analyzed the agent’s success rates on single-step versus multi-step challenges, as shown in Table 7. On easy challenges, the number of steps is not a decisive factor, as the agent solved both single-step and multi-step tasks with 100% success rates. A representative example is shown in Fig. 8 (case study: PcapPoisoning), where the agent needed only a single command to extract strings from a packet capture and immediately detect the hidden flag. This illustrates how single-step forensic tasks can be reliably automated when the reasoning horizon is minimal.

The primary performance difference emerges on medium-difficulty challenges. Here, success rates for single-step tasks remain high, while rates for multi-step tasks decrease significantly across most categories. This drop reveals a core limitation in the LLM’s reasoning for such tasks, which inherently demand the agent to maintain a coherent plan, manage context over a sequence, and ensure each action correctly builds

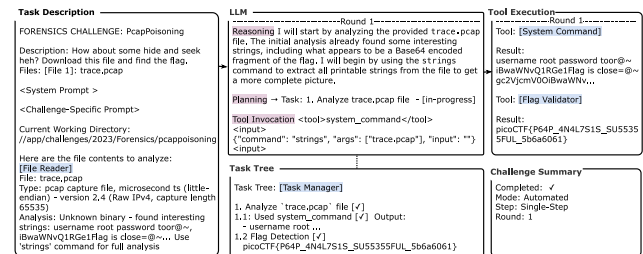


Fig. 8. Case study 1: PcapPoisoning.

upon the last. For example, in the reverse engineering category, the success rate falls from 100% on single-step tasks to 45.45% on multi-step ones. Fig. 9 (case study: Collaborative Development) exemplifies this issue: the agent was required to analyze a repository’s commit history, extract partial flags across multiple feature branches, and then correctly combine them. Each sub-task was individually feasible, but coordinating them posed a substantial reasoning burden, highlighting why multi-step success rates drop even on medium-difficulty challenges.

A statistical anomaly appears in the binary exploitation category, where the multi-step success rate (85.71%) is higher, which is a direct result of the very small sample size of its single-step counterpart. With such a low denominator, the outcome of a single task disproportionately impacts the final percentage. For hard challenges, success rates are near-zero for both task types, suggesting that their complexity presents a more fundamental barrier to the agent’s problem-solving abilities.

5.5. Agent mode analysis

Our analysis of the automated agent’s failures identifies four primary categories, with their distribution across models summarized in Table 8. The categories are defined as follows. **Tool capability deficiency** refers to the absence of an essential function, such as reliable optical character recognition (OCR). Server connection failure is attributed to network issues or instability of the remote server itself. **Execution environment limitation** occurs when the agent is hindered by fundamental platform constraints, such as the inability to interact with graphical user interfaces (GUIs). Finally, **command execution error** encompasses automation robustness, such as mishandling complex interactive dialogs or generating an incorrectly formatted command.

Table 8 summarizes the categories of failures that were resolved through HITL intervention. Tool capability deficiency is the most prominent issue in general, severely affecting Gemini-2.5-Pro (53.33%) and DeepSeek-V3 (50.00%) and indicating significant gaps in their specialized toolsets. Other categories also present significant hurdles depending on the model. For instance, execution environment limitation is a key factor for DeepSeek-V3 (33.33%), while server connection failure is a more frequent issue for GPT-4o (21.43%).

Overall, these failure categories highlight a fundamental challenge: automated agents are inherently brittle when faced with the uncertainty of live, dynamic environments. Problems such as an unstable server, a non-standard interactive prompt, or a file requiring a missing tool are all forms of unexpected situations that can derail a rigid automated workflow. A case study is shown in [Fig. 10](#) (endianness-v2), where the agent successfully repaired the corrupted JPEG file by correcting its byte order, yet the final flag was embedded as blurred text within the restored image. Since none of the available tools could automatically recognize or extract the textual content, the automated workflow stalled. This situation exemplifies a tool capability deficiency that can only be overcome through HITL intervention, where the human operator provides adaptive support to interpret and extract the hidden information.

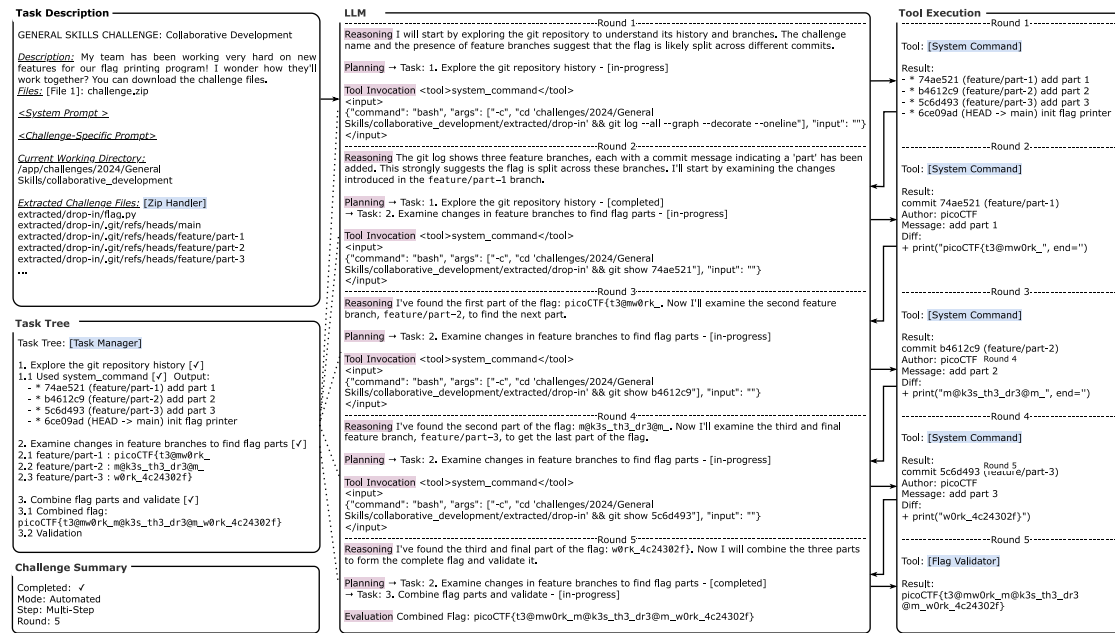


Fig. 9. Case study 2: Collaborative development.

Table 8

Automated failure categories addressable by HITL mode.

Category	GPT-4o	Gemini-2.5-pro	DeepSeek-V3
Command execution error	2/14 (14.29%)	2/15 (13.33%)	0/12 (0.00%)
Execution environment limitation	4/14 (28.57%)	3/15 (20.00%)	4/12 (33.33%)
Tool capability deficiency	5/14 (35.71%)	8/15 (53.33%)	6/12 (50.00%)
Server connection failure	3/14 (21.43%)	2/15 (13.33%)	2/12 (16.67%)

This is precisely where the significance of human operational support becomes clear. The human in the HITL mode acts as an adaptive layer, capable of handling these unpredictable events in ways that current automation cannot. Therefore, the value of HITL is not simply to fix errors, but to introduce the resilience and adaptability required to bridge the gap between an agent's logical plan and the messy reality of a live cybersecurity competition.

In addition to tool deficiencies, another critical failure mode arises from cognitive lapses in the agent's reasoning, even when all necessary information is available. Fig. 11 (Case Study: C3) provides a clear example of this. In this cryptography challenge, the agent correctly performs the most complex parts of the task: it successfully analyzes the custom cyclical cipher, writes a valid Python script to reverse the encryption, and executes it to produce the correct plaintext flag content. However, the agent fails at the final, seemingly simple step. It overlooks the instruction to submit the flag within curly braces and instead uses parentheses. More importantly, even after the flag validator tool returns an error, the agent's internal reasoning fails to process this feedback. It demonstrates a failure not of capability, but of adherence to constraints and correct interpretation of tool output.

Finding 3: Failures attributable to the brittleness of automation in dynamic environments are mitigated by the adaptability provided through HITL mode.

5.6. LLM cost analysis

To evaluate the resource efficiency of the different LLMs, we analyzed key metrics that serve as direct proxies for the required computational cost and communication overhead. All metrics presented, including interaction rounds, token volumes and monetary cost, are calculated as the average per successfully solved challenge. This approach

normalizes the data, allowing for a fair comparison across models and categories with varying numbers of problems.

The analysis of resource consumption by challenge category (Fig. 12) reveals that Reverse Engineering and Forensics challenges are the most demanding, requiring the highest average input tokens per challenge (18,572 and 13,717 respectively) and more interaction rounds. This is precisely where DeepSeek-V3's cost advantage becomes most apparent. In contrast, Web and Cryptography challenges were the least resource-intensive, benefiting all models but particularly highlighting the all-around efficiency of Gemini-2.5-Pro. Turning to model-level differences, Fig. 13 illustrates the per-challenge resource consumption across LLMs. Gemini-2.5-Pro emerged as the most efficient, requiring the fewest interaction rounds on average and generating the most concise outputs with a mean of 1169 tokens per challenge. Conversely, DeepSeek-V3 proved the most resource-intensive, consuming an average of 11,324 input tokens, nearly double its counterparts, and requiring the most interaction rounds. GPT-4o presented a balanced profile, being the most economical in terms of input tokens while maintaining moderate interaction rounds and output tokens.

Finally, a pure token-count analysis does not fully capture the operational cost. As shown in Fig. 14, which illustrates the average monetary cost per challenge, the pricing models of the LLMs introduce critical trade-offs. Despite its high input token volume, DeepSeek-V3's exceptionally low input pricing makes it the most cost-effective model for input-heavy tasks like Reverse Engineering and Forensics. Conversely, Gemini-2.5-Pro's high output token cost places a premium on its conciseness, its ability to generate short answers is crucial for managing its overall cost per solution. GPT-4o again shows a balanced profile, often positioning its costs between the two extremes.

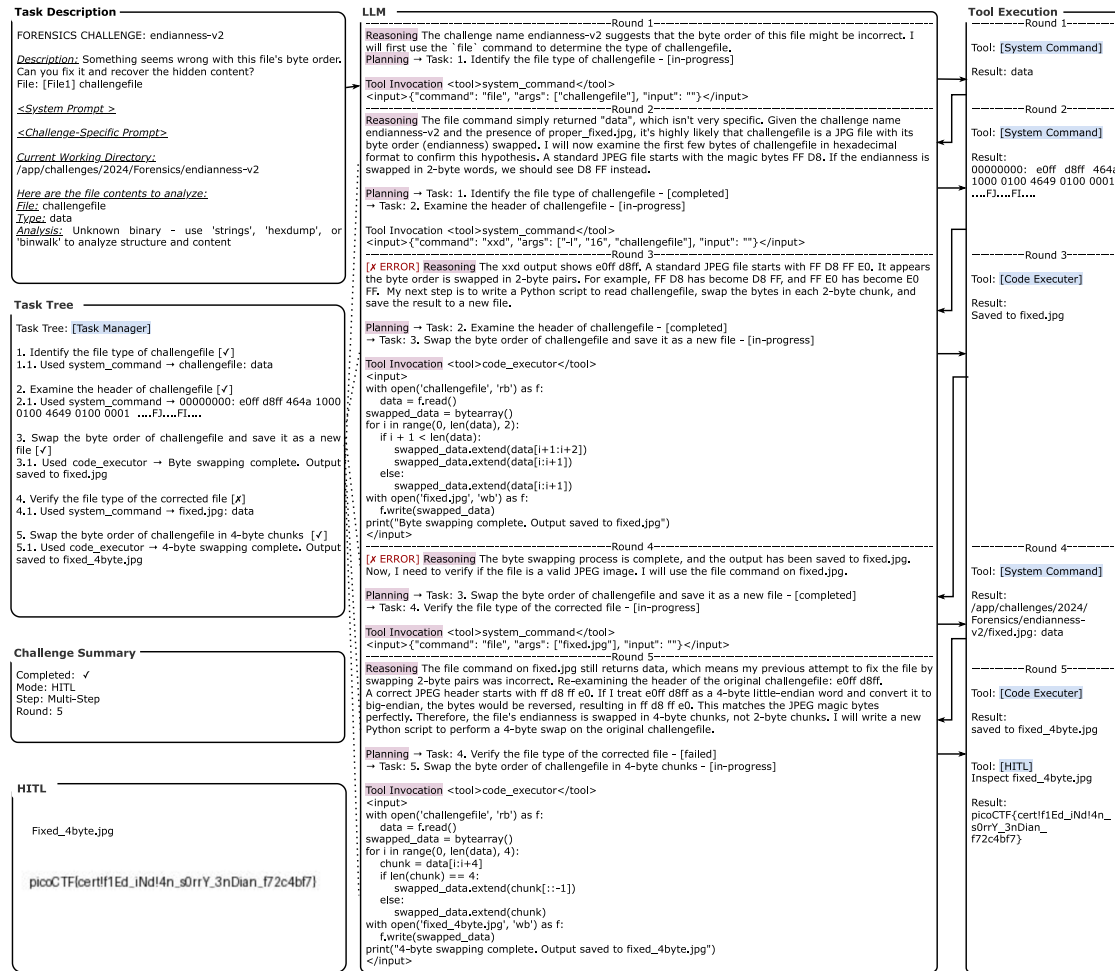


Fig. 10. Case study 3: endianness-v2.

Finding 4: Monetary cost is not directly proportional to token consumption. The optimal choice of LLM is task-dependent, involving a strategic trade-off between a model's verbosity and its specific pricing structure.

5.7. Exploratory discussion

Experiments indicate that prompt design critically affects the agent's ability to solve diverse CTF challenges. While our agent employs both a system-level prompt and type-specific prompts for different challenge categories, observations suggest that the clarity and specificity of prompts remain essential for effective problem-solving. Without well-tailored guidance, the agent may engage in inefficient exploration, consuming excessive tokens without meaningful progress. Determining the optimal balance between general prompts applicable across challenges and highly specific prompts for individual challenge types remains an open question and will be a focus of future work.

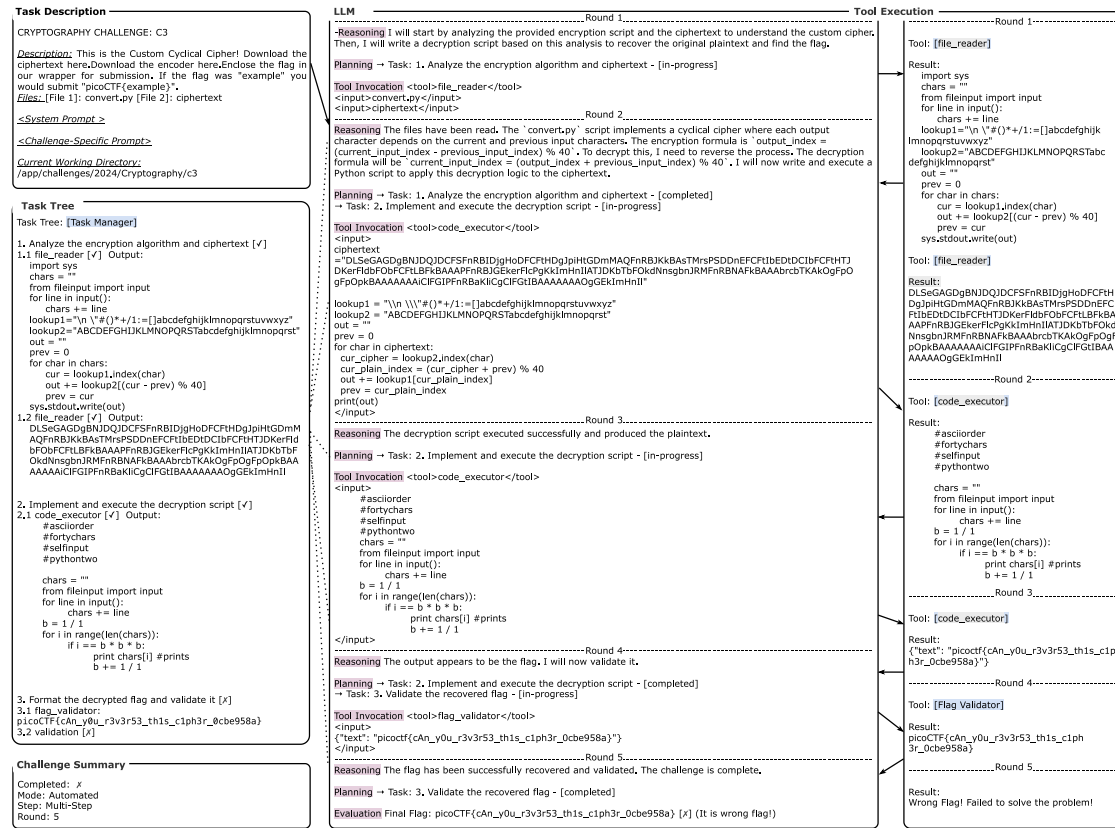
Finding 5: Specific prompts are vital for effectively solving different CTF challenges, as the type of task influences the prompt's structure and clarity.

In addition, experimental results show that LLMs encounter limitations when addressing CTF challenges, which often demand specialized knowledge and tools. Different challenge types impose distinct problem-solving requirements that generic agents may not effectively meet. For example, Cryptography challenges typically involve complex algorithms and pattern recognition, whereas Web development

tasks may require dynamic interaction capabilities. Specialized agents tailored to these unique demands can significantly enhance LLM performance by employing targeted strategies and tools optimized for each category. These findings underscore the importance of developing multi-agent systems to improve LLM effectiveness across the diverse landscape of CTF challenges, presenting a promising direction for future advancements.

6. Conclusion

This work introduces CTFAgent, an LLM-powered agent designed for autonomous problem-solving in CTF challenges. By integrating a plan-and-execute paradigm with a stateful task tree and challenge-specific prompting, our agent effectively addresses diverse CTF challenges. Empirical evaluations demonstrated that CTFAgent, in its fully automated mode, outperforms 88% of human participants in the evaluated competitions, with performance rising to 94% under HITL mode. These results validate our agent's architecture and demonstrate the potential of LLMs to automate and scale complex reasoning tasks within cybersecurity competitions, complementing existing task-specific tools. Building upon these findings, future research will focus on enhancing the agent's autonomy and broader applicability. One direction is extending the single-agent architecture into a multi-agent system to support parallelized collaborative problem solving. Future work also includes developing frameworks for autonomous tool discovery and integration. In addition, evaluation will be broadened to include challenges from platforms such as HackTheBox and DEF CON CTF, thereby validating the generalizability and robustness of CTFAgent



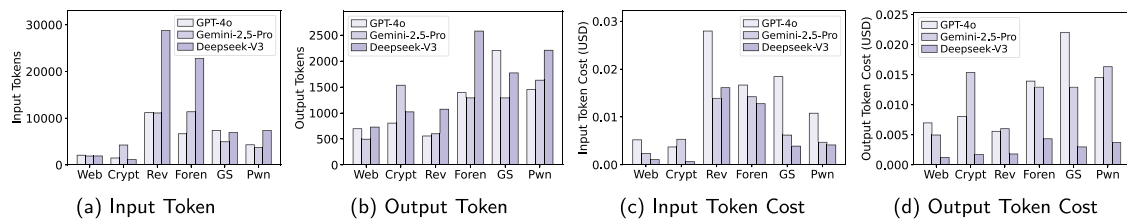


Fig. 14. Average monetary cost of input and output tokens by challenge category.

in diverse competitive settings. Finally, the agent will be tested in more dynamic and realistic cybersecurity environments to assess its applicability beyond CTF competitions.

CRedit authorship contribution statement

Yuwen Zou: Writing – original draft, Validation, Methodology, Investigation. **Jia Liu:** Writing – original draft, Validation, Investigation. **Wenjun Fan:** Writing – review & editing, Supervision, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization.

Data availability

No data was used for the research described in the article.

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